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FACULTY OF COMPUTER SCIENCE & MATHEMATICS

WEB-BASED APPLICATION DEVELOPMENT
CSE3023

SMART SELF-SERVICE LAUNDRY SYSTEM

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SEMESTER 2 2025/2026

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1.0 Project Overview

The Smart Self Service Laundry System is a web-based application that improves the convenience of laundry services to the customers. The system enables users to control their laundry digitally. This system is designed to provide a more convenient and faster self-service laundry experience. The system is a digital platform that helps the users easily find the nearby laundry stores and compare their availability, making it easier to choose the most convenient one. The application combined with real-time machine status avoids unnecessary waiting time and improves overall efficiency in planning their laundry tasks.

After a customer selects a store, they can check if there are washers or dryers available. If the selected store is full, users are recommended to look for other nearby options, through the system. Customers can come directly to the store where machines are available. There is no provision for reservations in the system. This methodology guarantees fair usage and promotes real-time decision making based on the availability of the real machine.

After loading their laundry, users can make payments conveniently using digital methods such as QR code or online banking. The system then automatically activates a timer linked to the selected machine, allowing customers to leave the premises and manage their time more freely. Once the laundry cycle is complete, a notification is sent to alert users that their items are ready for collection. After unloading, the machine status is updated in the system, making it available for the next customer.

Apart from the basic functionality, the system enhances user engagement via a review and reward mechanism. Customers can rate their experience and provide feedback to support future improvements. It also has a loyalty points system. Users get points according to their spendings which can be redeemed later as discounts during future transactions. All in all, the application aims to provide a convenient, efficient and user-friendly solution to modern self-service laundry needs.

2.0 Problem Statement

Traditional self-service laundromats still rely heavily on manual operations, which limits efficiency and negatively impacts customer experience. Customers currently have no way to check machine availability in real time before arriving, often resulting in long waiting times, overcrowding, and frustration. This lack of visibility reduces convenience, especially for users with tight schedules such as students and working individuals.

In addition, many laundromats continue to depend on cash-based transactions, which are increasingly inconvenient in today's digital and cashless society. Customers may face difficulties such as not having exact change or needing to locate nearby cash machines, further reducing service accessibility and satisfaction.

From the management perspective, the absence of a centralized digital system makes it difficult for laundromat owners to efficiently monitor operations. Machine usage, transaction records, and customer activity are often not tracked systematically, leading to poor decision-making, inefficient maintenance scheduling, and potential revenue loss. Without proper data insights, owners are also unable to optimize resource allocation or identify peak usage periods.

Furthermore, the lack of automation and integration in traditional laundromats limits scalability and modernization. As technology continues to evolve, failing to adopt digital solutions may cause these businesses to fall behind competitors that offer smarter, more user-friendly services.

Therefore, there is a strong need for a Smart Self-Service Laundry System that provides real-time machine availability, supports cashless payments, and enables centralized management. Such a system aims to improve operational efficiency, enhance user convenience, and support the digital transformation of traditional laundromat services.

3.0 Use Case Diagram



Figure 2.0.1 Use Case Diagram for Smart Self Service Laundry System

1. Actors

- Customer: The end-user who use the system to do laundry in their area.

2. Key Use Cases

- Registration and Login: User creates a new account and stores the user information. After registration, user can access the system and login for next use.
- Get Location and Check Machine Availability: Able the user to check availability of any machines at nearby laundry.
- Make Payment and Get Transaction: Perform cashless payment and the transaction will be recorded.
- Get Timer and Manage Notification: User will get notification about their laundry progress tracking.
- Manage Loyalty and Discount: Earn points based on laundry usage. User will be rewarded with discount.
- Manage Review: User can submit feedback and give ratings so future improvement for the system can be made.

4.0 Class Diagram

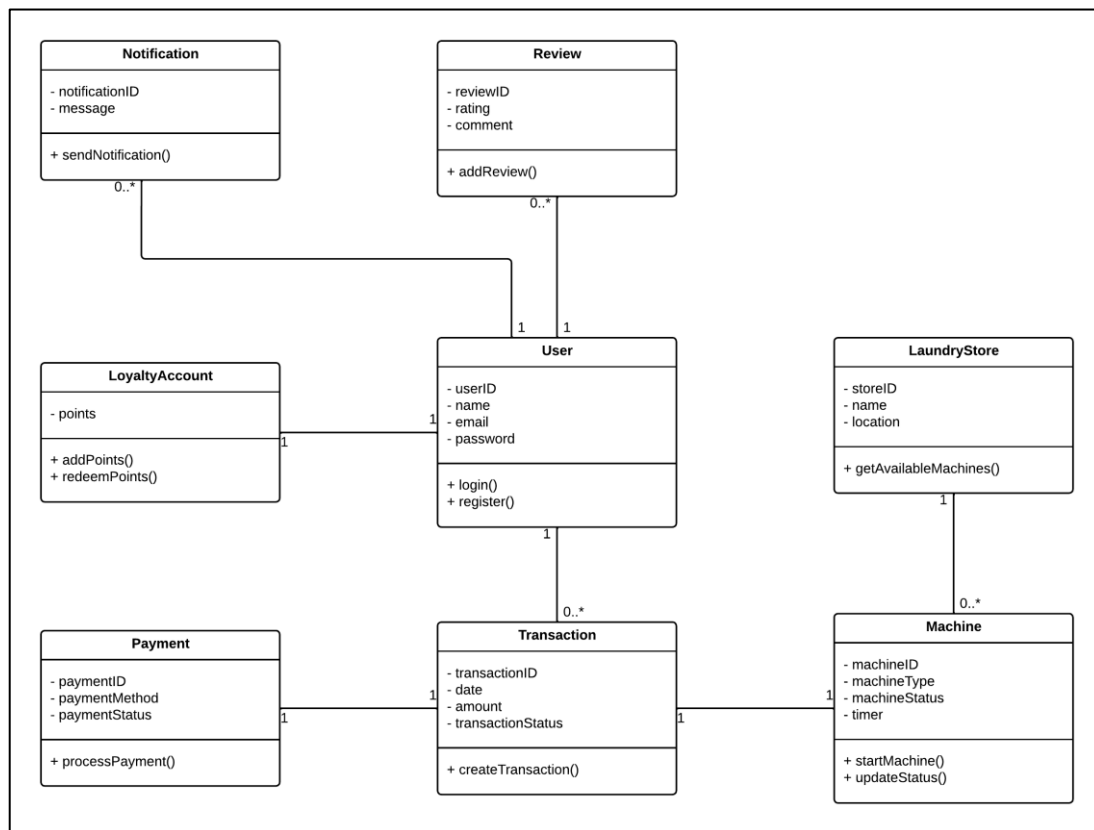


Figure 3.0.1 Class Diagram for Smart Self Service Laundry System

The relationship between classes:

1. One User can make many Transaction
2. One LaundryStore has many Machine
3. Each Machine belongs to one LaundryStore
4. One Transaction linked to one Machine and one User
5. One Payment belongs to one Transaction
6. One User has one LoyaltyAccount
7. One User can give many Review
8. One User can receive many Notification

Flow of the system:

1. User selects desired store
2. System shows machines available
3. User go to the selected store and selects machine
4. User makes payment via online banking or QR payment.
5. Transaction created
6. Machine starts timer
7. System send notification if the machines has stopped and ready for pick up
8. After transaction done, machine automatically become available again in the system
9. User earns points, and can give reviews

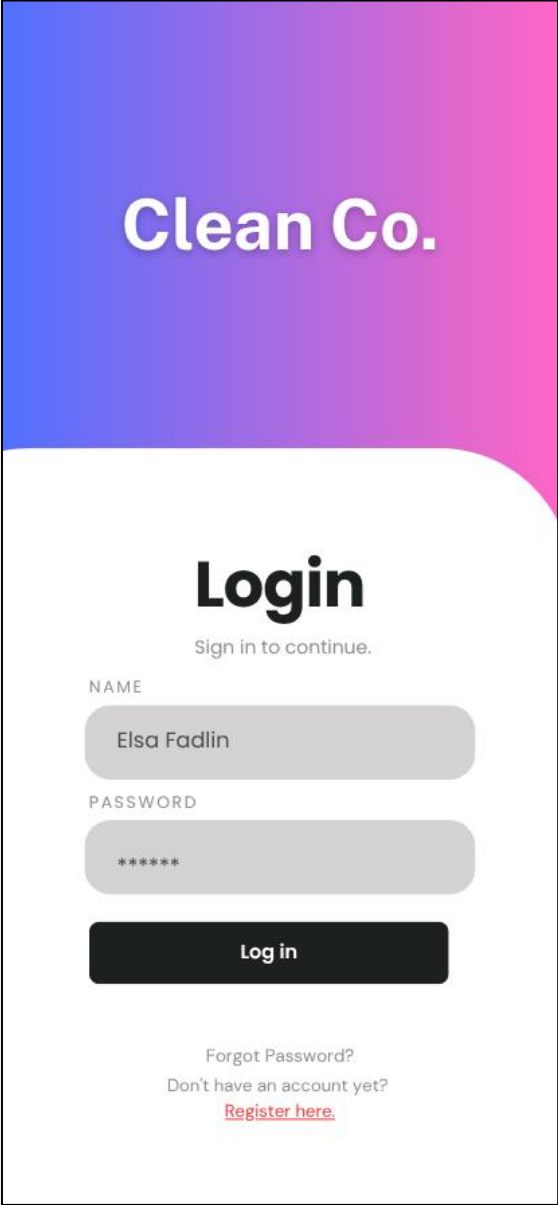
5.0 Module of each member

The Smart Self-Service Laundry System is designed with several functional modules that work together to manage the overall operation efficiently. Each module is assigned to a team member and focuses on a specific function. The Timer and Notification module by Elsa Fadlin manages laundry timing and user alerts, the Payment and Transaction module by Nik Mariyah Sofeah handles payments and transaction records, and the Location and Machine Availability module by Siti Aisyah Arwiah provides information on machine availability. These modules help users monitor activities, make payments, and find available machines easily, while ensuring all information is managed effectively using CRUD (Create, Retrieve, Update, Delete) operations.

Name	Module	CRUD Functions
Elsa Fadlin	Timer and Notification	Create The system creates a new timer when a user starts a session and generates notifications such as reminders or alerts. Retrieve Users can view active timers and past notification history stored in the system. Update The system updates the timer status (e.g., running, paused, completed) and modifies notification settings or schedules. Delete Expired timers and old notifications can be removed from the system to maintain efficiency and storage.
Nik Mariyah Sofeah	Payment and Transaction	Create A new transaction record is created when a user makes a payment. Retrieve Users and can view transaction history, payment status. Update Payment status can be updated (e.g., pending, completed, failed).

		Delete Invalid or duplicate transaction records may be removed from the system.
Siti Aisyah Arwiah	Location and Machine Availability	Create New machine records and locations are added into the system. Retrieve Users can check available machines and their locations in real time. Update Machine availability status is updated (e.g., available, in use, under maintenance). Delete Outdated or removed machines can be deleted from the system.

6.0 UI Design

The image shows a mobile app login screen for 'Clean Co.'. The top half of the screen has a blue-to-pink gradient background with the company name 'Clean Co.' in white. The bottom half is white. The login form is centered and includes a title 'Login', a subtitle 'Sign in to continue.', a name input field with the text 'Elsa Fadlin', a password input field with masked characters '*****', a black 'Log in' button, and links for 'Forgot Password?' and 'Don't have an account yet? Register here.'.

Clean Co.

Login

Sign in to continue.

NAME

Elsa Fadlin

PASSWORD

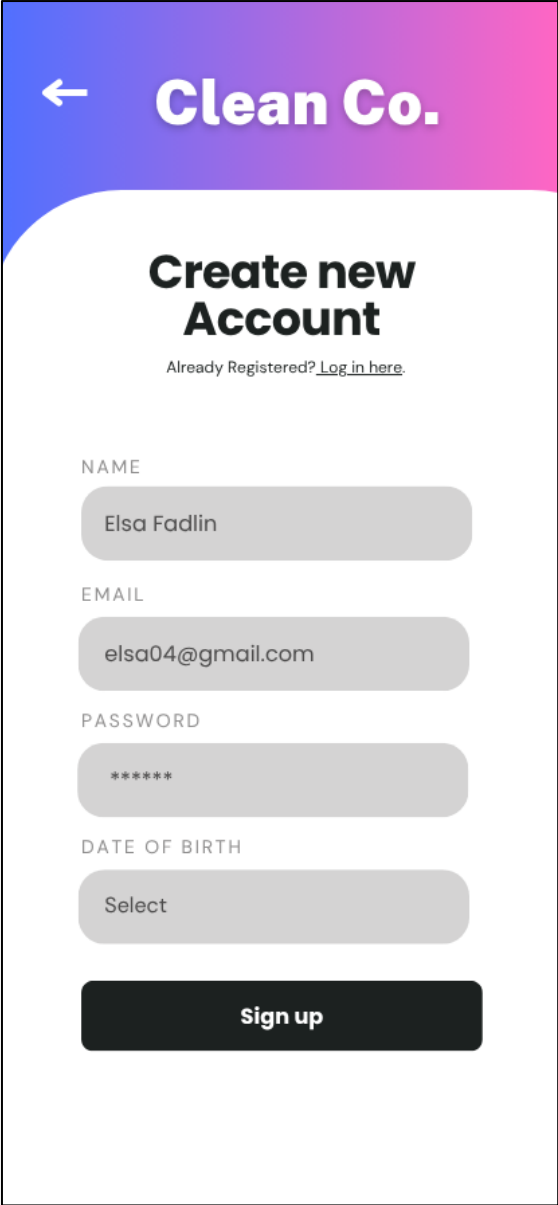
Log in

Forgot Password?

Don't have an account yet?

[Register here.](#)

Figure 6.0.1 Login UI Design

The image shows a mobile app interface for creating a new account. At the top, there is a header bar with a blue-to-pink gradient. On the left of the header is a white back arrow, and on the right is the text "Clean Co." in white. Below the header, the main content area is white. It features the title "Create new Account" in bold black text. Underneath the title is a link that says "Already Registered? [Log in here.](#)". There are four input fields, each with a label above it: "NAME" with the value "Elsa Fadlin", "EMAIL" with the value "elsa04@gmail.com", "PASSWORD" with masked characters "*****", and "DATE OF BIRTH" with a dropdown menu showing "Select". At the bottom of the form is a black button with the text "Sign up" in white.

← Clean Co.

Create new Account

Already Registered? [Log in here.](#)

NAME

Elsa Fadlin

EMAIL

elsa04@gmail.com

PASSWORD

DATE OF BIRTH

Select

Sign up

Figure 6.0.2 Create Account UI Design

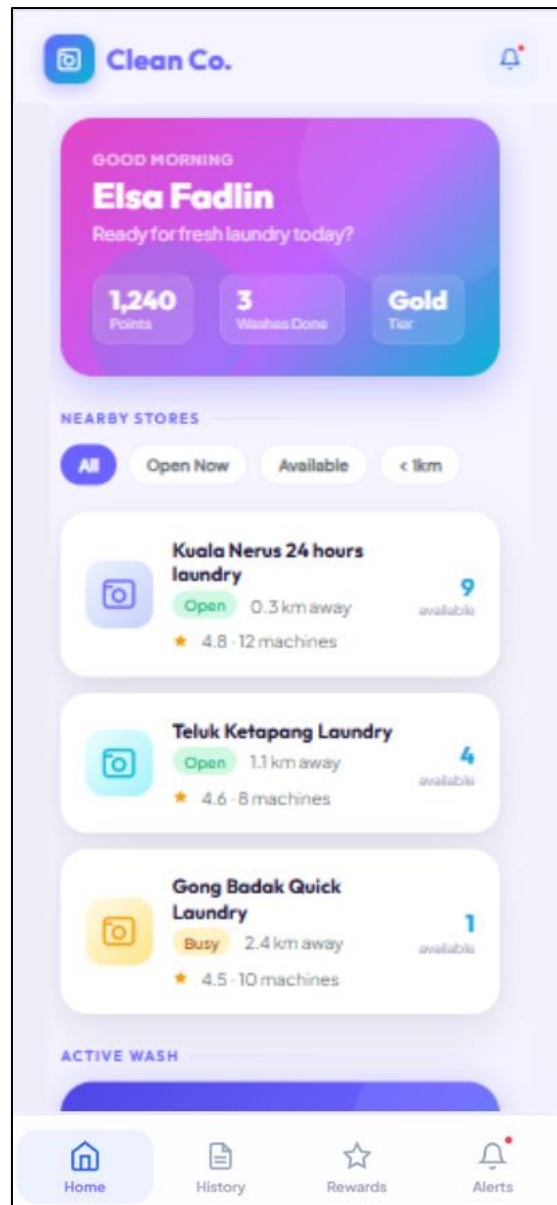


Figure 6.0.3 Home UI Design

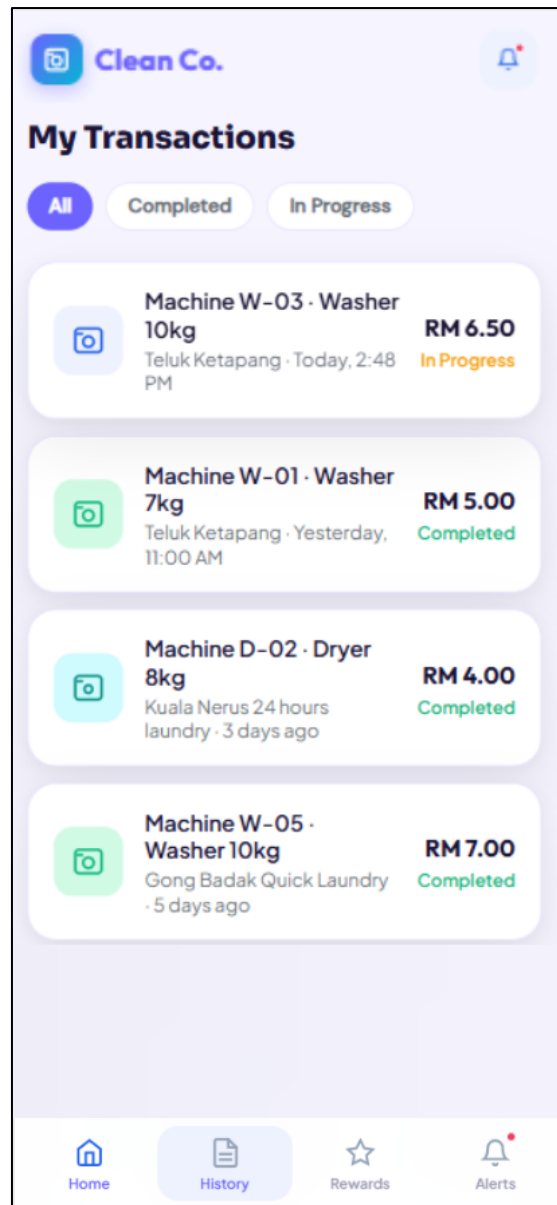


Figure 6.0.4 History UI Design

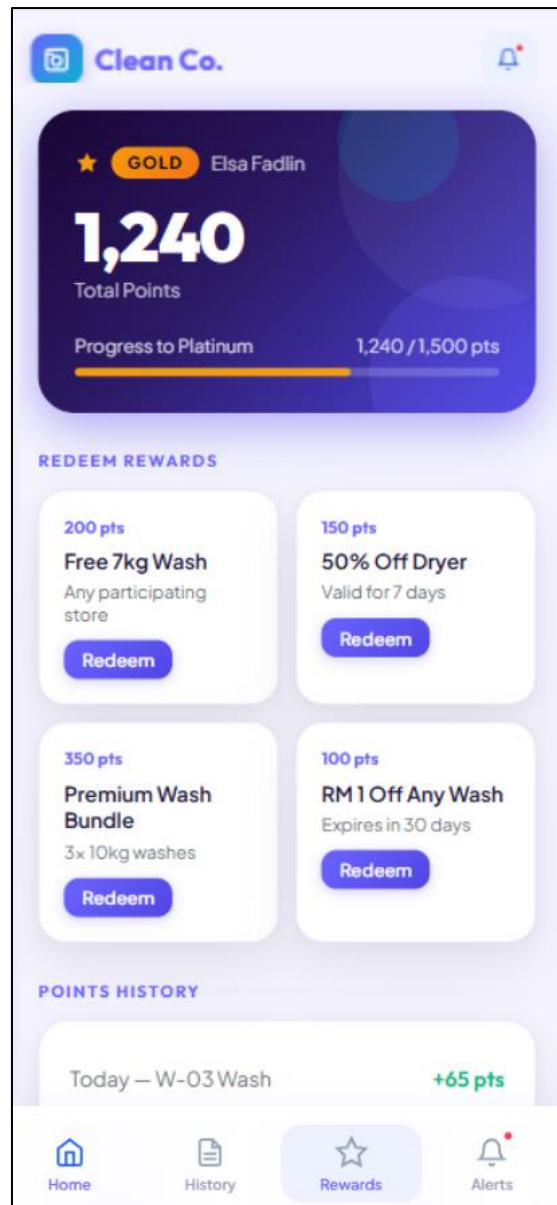


Figure 6.0.5 Reward UI Design

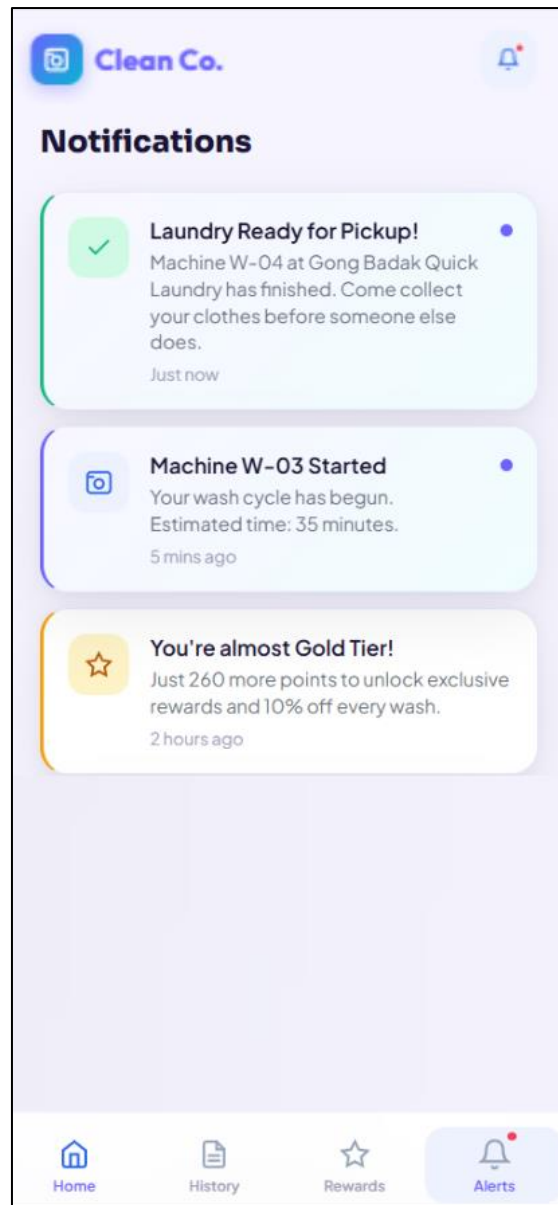


Figure 6.0.6 Alert UI Design

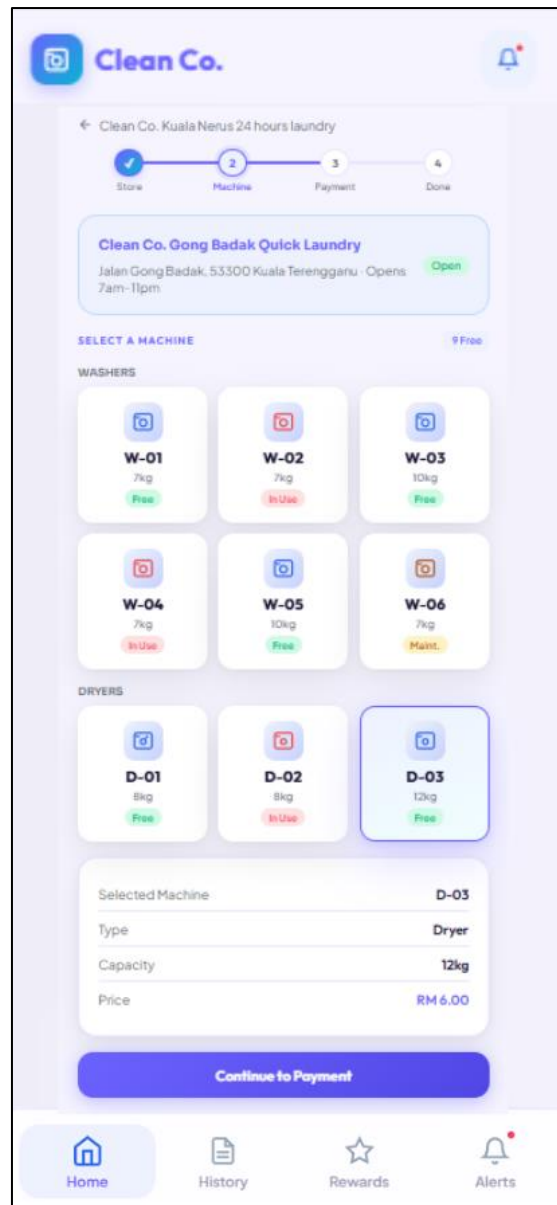


Figure 6.0.7 Select Store and Machine UI Design

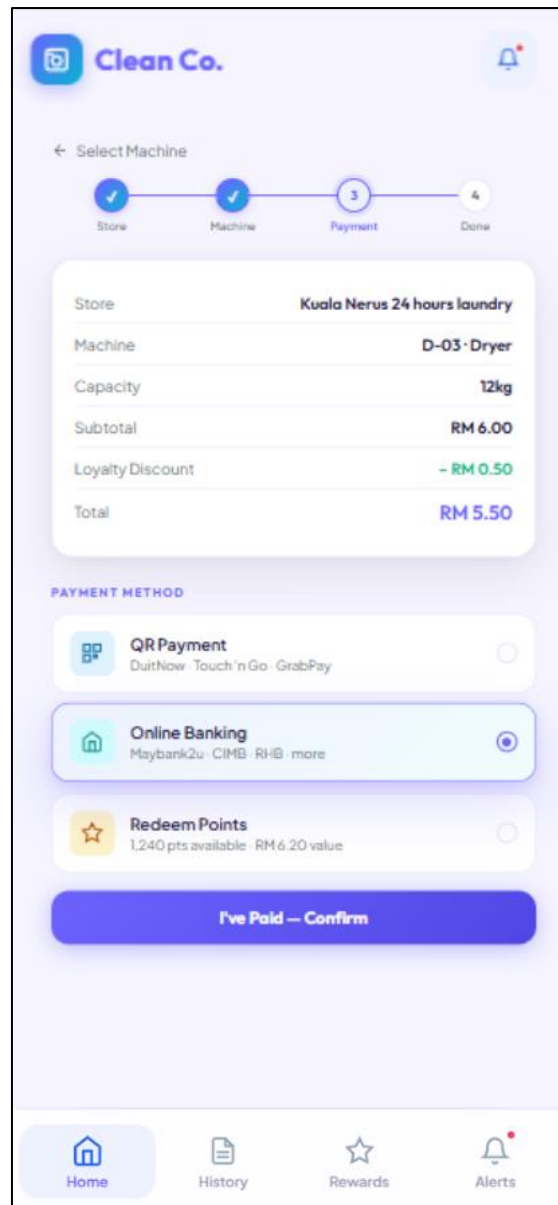


Figure 6.0.8 Payment UI Design

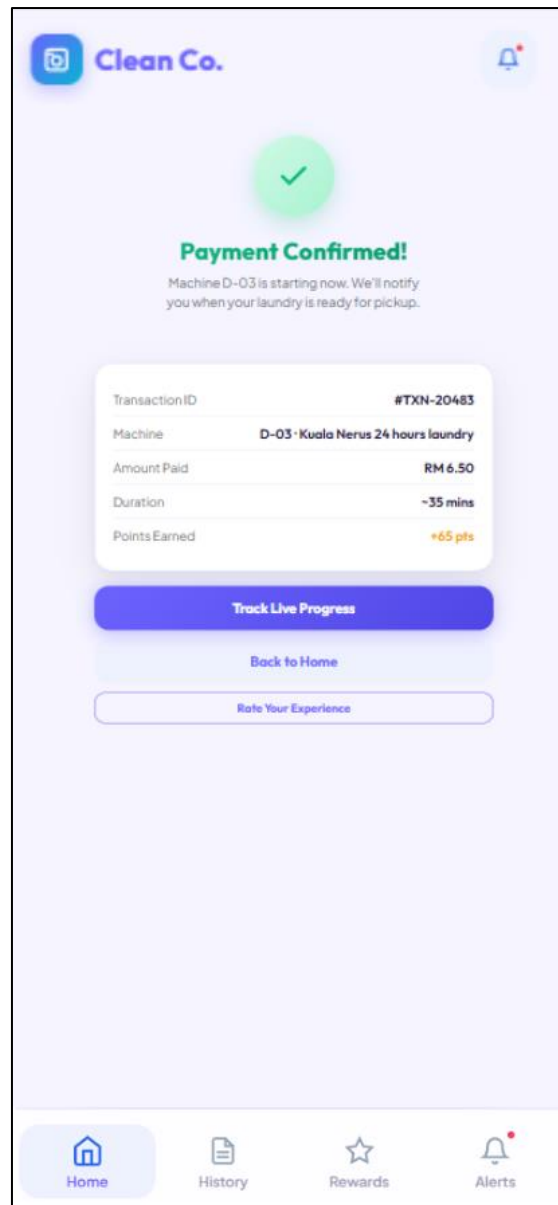


Figure 6.0.9 Payment Confirmed UI Design

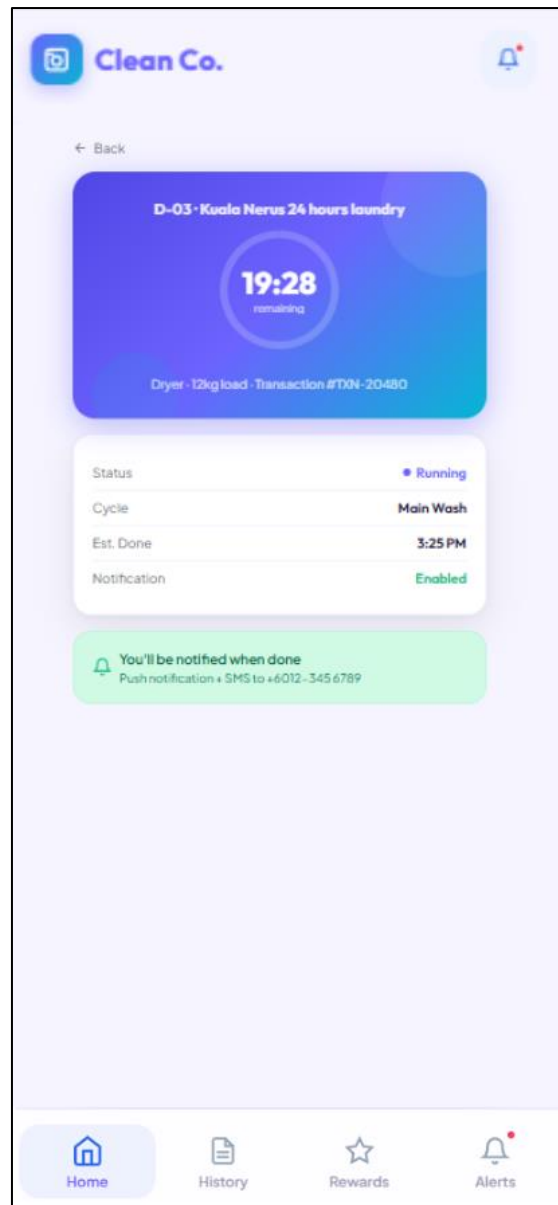


Figure 6.0.10 Timer Tracking UI Design

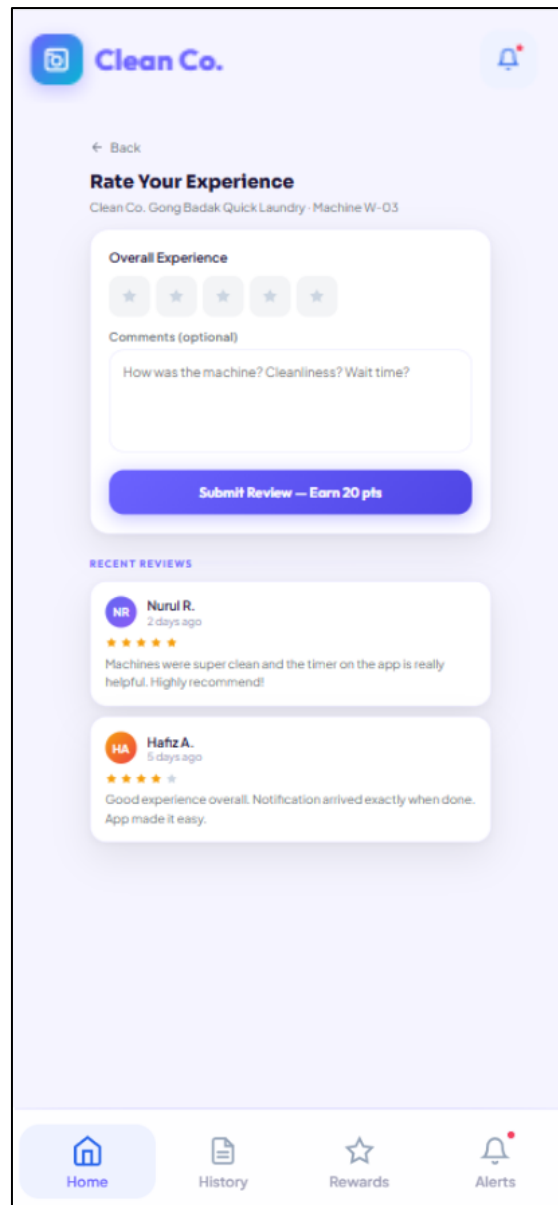


Figure 6.0.11 Review UI Design